

AWS Cloud Quest

Cloud Practitioner Track



Completion Summary

The AWS Cloud Quest – Cloud Practitioner Track was completed on the AWS Skill Builder platform. This program follows a game-based, hands-on learning approach to introduce foundational cloud computing concepts and core AWS services. Through interactive quests and real-world lab scenarios, the track focuses on practical understanding rather than theory alone.

Completion Highlights:

- Enrolled and actively participated in the AWS Cloud Quest – Cloud Practitioner Track
- Successfully completed all assigned quests and hands-on labs
- Achieved 100% completion status in the learning journey
- Earned the **AWS Cloud: Cloud Practitioner – Digital Badge**, validating foundational AWS knowledge

Credential Evidence:

AWS Cloud: Cloud Practitioner – Digital Badge

<https://www.credly.com/badges/0a44e013-317d-440a-ade9-a81e98d3350a>

Assignment 1: Cloud Computing Essentials

Aim

To understand the fundamentals of cloud computing and implement a simple, reliable web hosting solution using Amazon S3 static website hosting.

Description

This assignment focused on introducing basic cloud concepts and demonstrating how AWS enables scalable and cost-effective hosting. A static website was hosted using Amazon S3 by enabling static website hosting and configuring bucket permissions securely using a bucket policy. The task demonstrated how cloud storage services can be used to host web content without managing servers.

Key Learning

- Understanding cloud-based hosting models
- Working with Amazon S3 buckets
- Configuring static website hosting
- Applying bucket policies for secure access

Screenshot



Assignment 2: Cloud First Steps

Aim

To deploy compute resources using Amazon EC2 and understand the concept of Availability Zones for high availability.

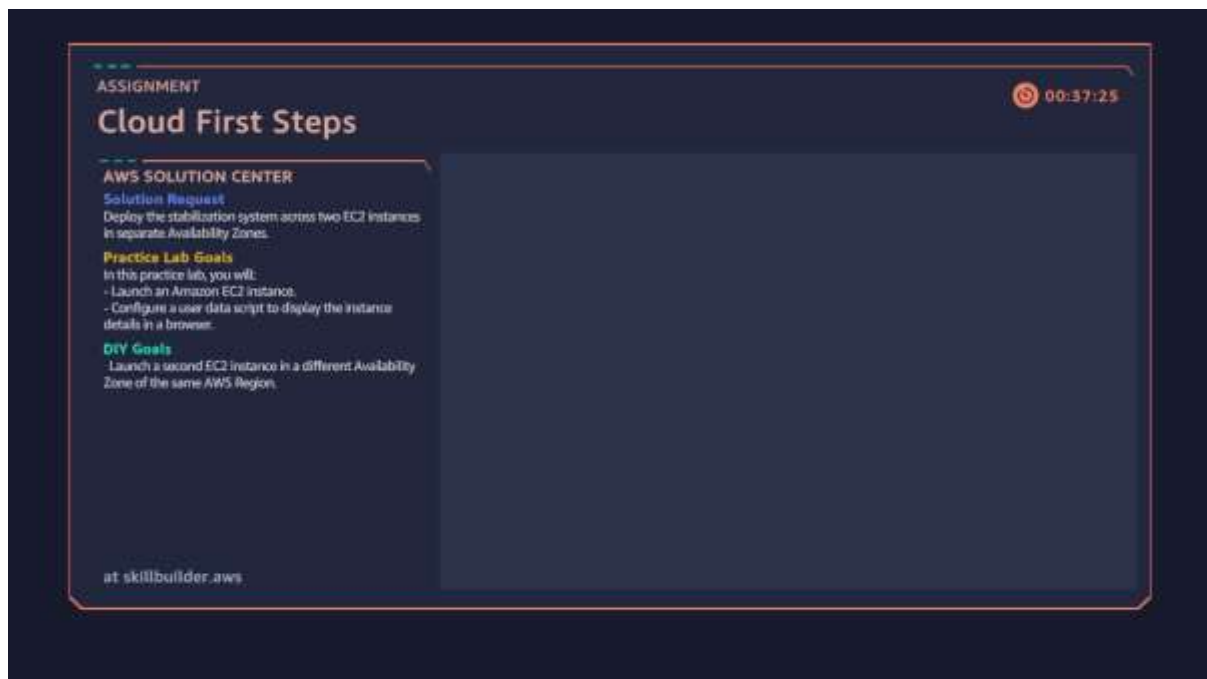
Description

In this assignment, an Amazon EC2 instance was launched using a selected AMI and instance type. User data scripts were configured to display instance details in a browser. A second EC2 instance was launched in a different Availability Zone within the same region to understand fault tolerance and availability concepts.

Key Learning

- Launching EC2 instances
- Understanding regions and Availability Zones
- Using user data scripts
- Basic EC2 instance management

Screenshot



Assignment 3: Computing Solutions

Aim

To explore EC2 scalability and performance optimization by modifying instance types and accessing instance metadata.

Description

This assignment involved analyzing EC2 instance attributes and scaling the instance by changing its type to a larger general-purpose instance. The task also included connecting to the EC2 instance using AWS Systems Manager and viewing instance metadata through the browser, reinforcing compute scalability concepts.

Key Learning

- Understanding EC2 instance types
- Scaling EC2 instances vertically
- Using AWS Systems Manager
- Viewing EC2 instance metadata

Screenshot



Assignment 4: Networking Concepts

Aim

To understand AWS networking fundamentals by configuring VPC components and security controls.

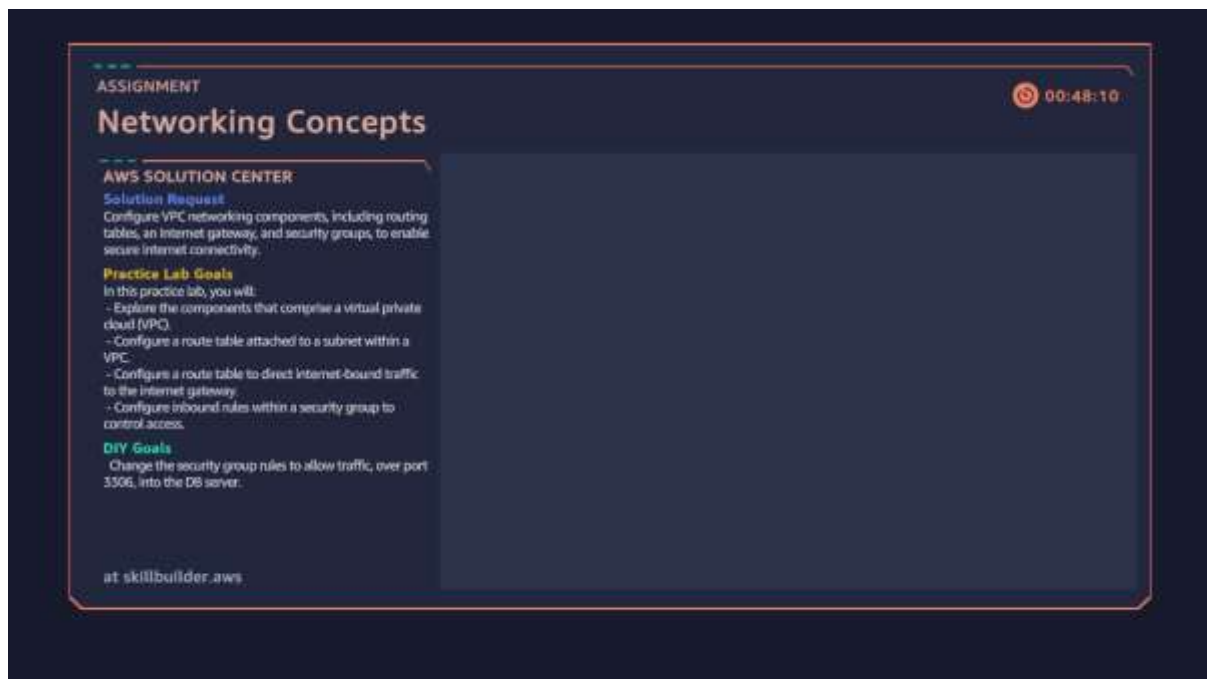
Description

This assignment focused on exploring Virtual Private Cloud (VPC) components including subnets, route tables, internet gateways, and security groups. Inbound rules were configured to allow traffic on a specific port to enable database connectivity, highlighting AWS network security best practices.

Key Learning

- Understanding VPC architecture
- Configuring route tables and gateways
- Security groups and inbound rules
- Network-level access control

Screenshot



Assignment 5: Cloud Economics

Aim

To understand AWS pricing models and estimate cloud costs using AWS pricing tools.

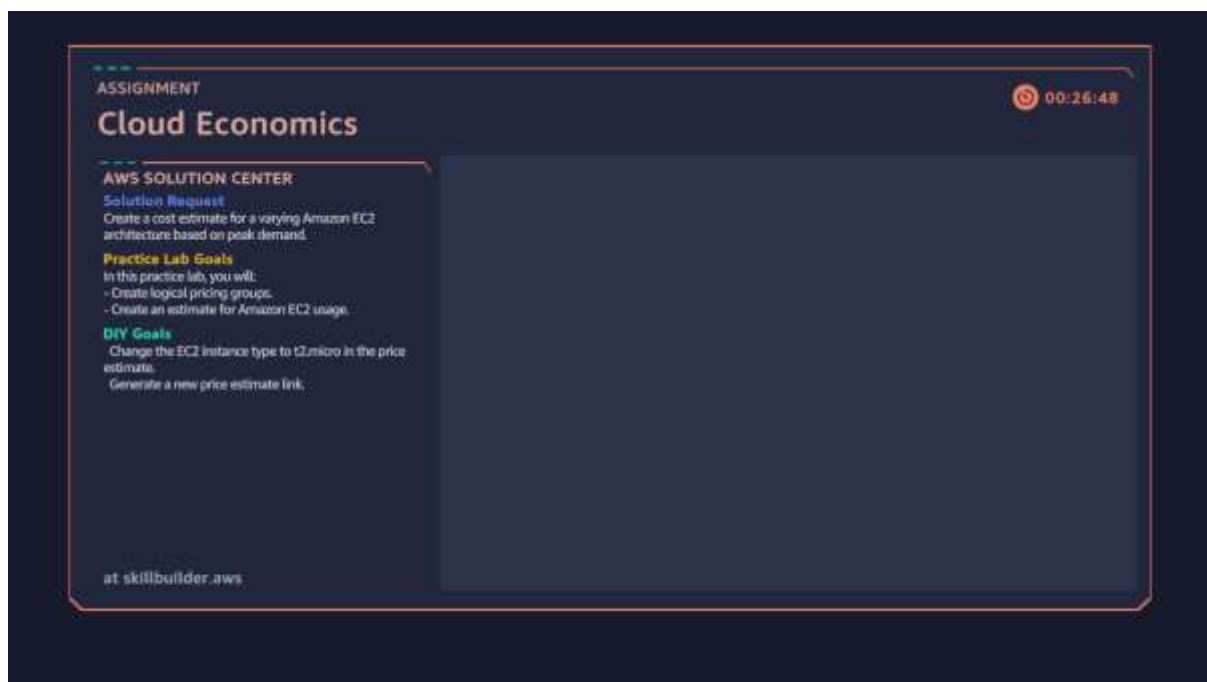
Description

This assignment introduced AWS cost management concepts by creating a pricing estimate for Amazon EC2 usage. Instance types were modified to analyze cost variations, and a new price estimate link was generated to understand demand-based pricing and cost optimization strategies.

Key Learning

- AWS pricing fundamentals
- EC2 cost estimation
- Cost optimization awareness
- Using AWS pricing tools

Screenshot



Assignment 6: Databases in Practice

Aim

To implement managed relational database solutions using Amazon RDS with high availability and backup features.

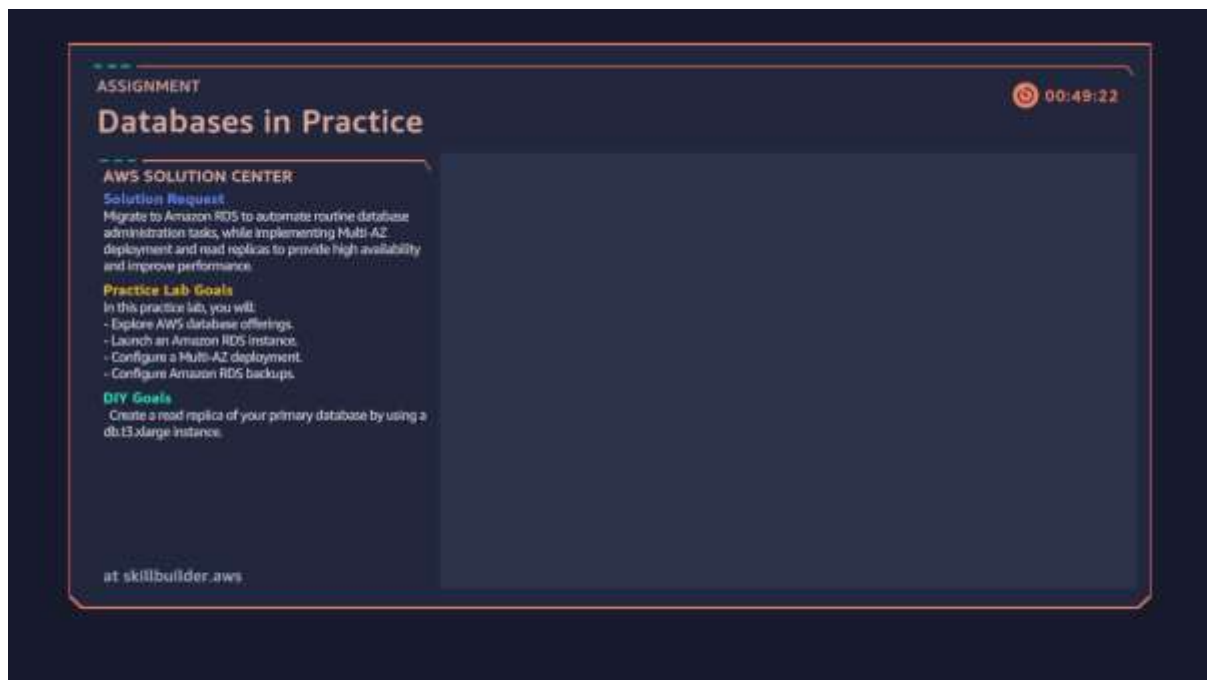
Description

This assignment focused on launching an Amazon RDS instance and configuring Multi-AZ deployment for fault tolerance. Automated backups were enabled, and a read replica was created to improve read performance and availability.

Key Learning

- Amazon RDS features
- Multi-AZ deployments
- Automated backups
- Read replicas for performance optimization

Screenshot



Assignment 7: First NoSQL Database

Aim

To create and interact with a NoSQL database using Amazon DynamoDB.

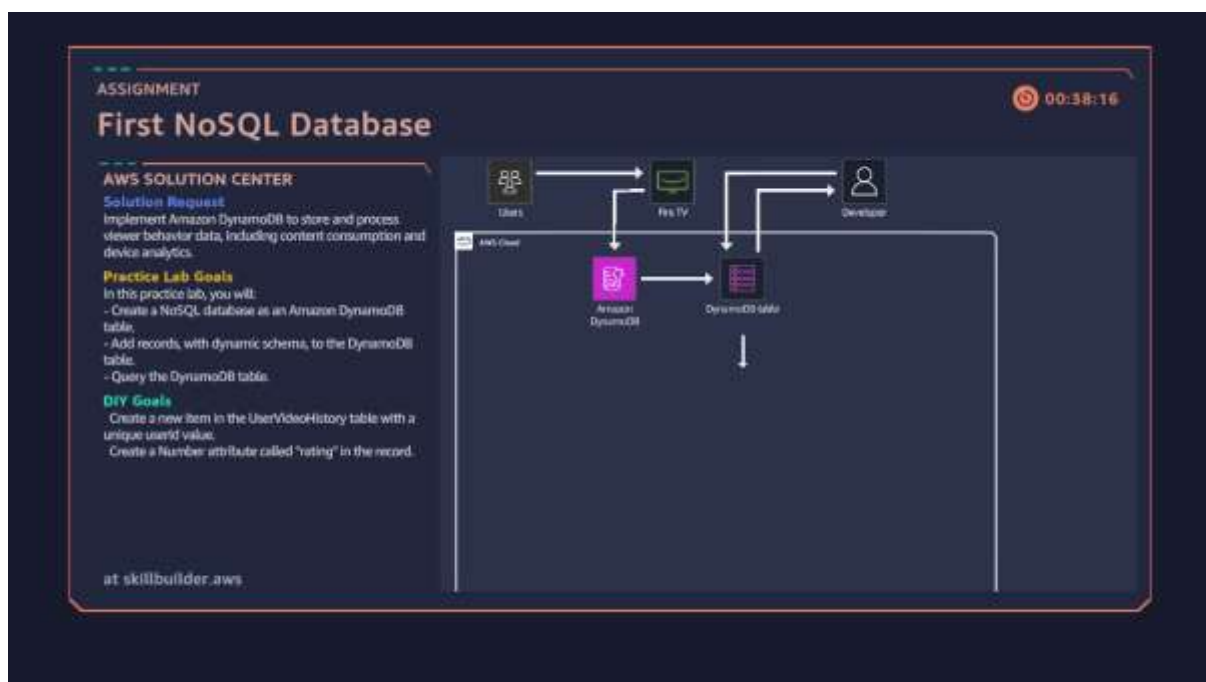
Description

In this assignment, an Amazon DynamoDB table was created to store user activity data. Records were added using a flexible schema, and queries were performed to retrieve data. This demonstrated DynamoDB's scalability and performance for real-time applications.

Key Learning

- NoSQL database concepts
- DynamoDB table creation
- Working with items and attributes
- Querying DynamoDB tables

Screenshot



Assignment 8: File Systems in the Cloud

Aim

To implement a shared, scalable file system using Amazon Elastic File System (EFS).

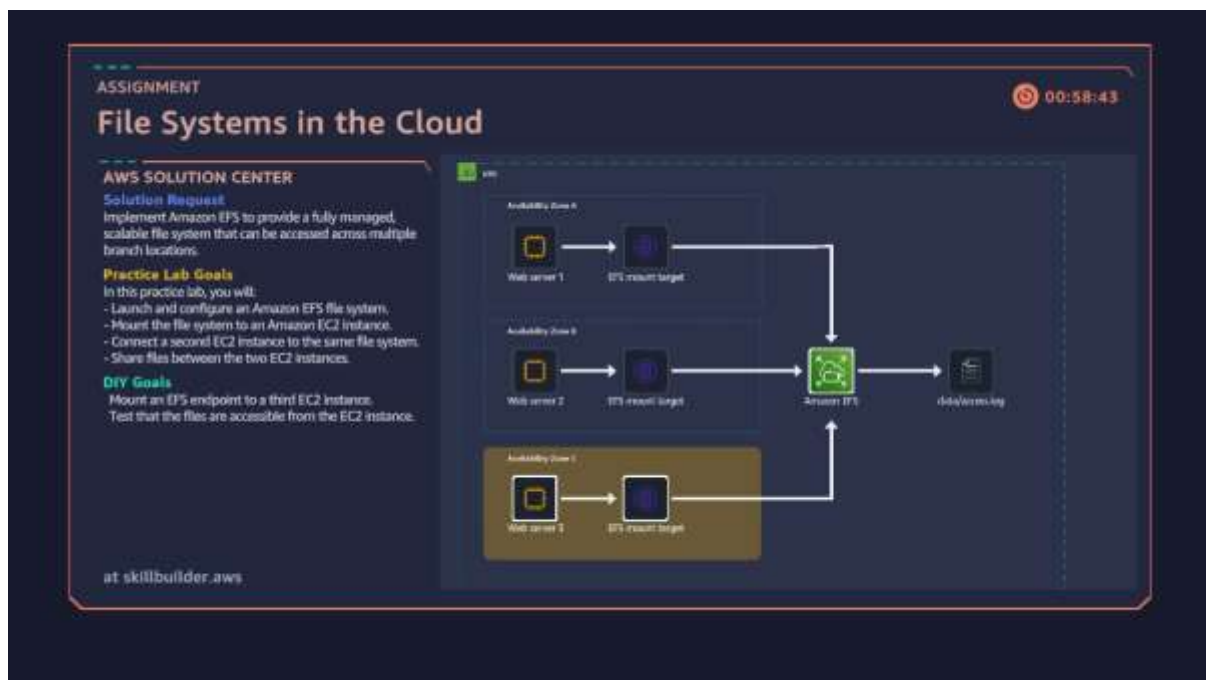
Description

This assignment involved creating an Amazon EFS file system and mounting it on multiple EC2 instances across different Availability Zones. Files created on one instance were accessible from other instances, demonstrating shared storage and high availability.

Key Learning

- Amazon EFS architecture
- Mount targets and Availability Zones
- Shared file systems
- Cloud-based storage scalability

Screenshot



Assignment 9: Auto-Healing and Scaling Applications

Aim

To implement automated scaling and self-healing mechanisms using Amazon EC2 Auto Scaling.

Description

This assignment focused on configuring an Auto Scaling group with a launch template and Amazon Machine Image (AMI). CloudWatch alarms were used to trigger scaling actions based on workload demand. Scheduled scaling was also configured to automatically reduce resources during low usage periods.

Key Learning

- Auto Scaling group configuration
- Launch templates and AMIs
- CloudWatch alarms
- Automated scaling and self-healing

Screenshot



Assignment 10: Highly Available Web Applications

Aim

To deploy a highly available web application using multiple Availability Zones and an Application Load Balancer.

Description

In this assignment, an Auto Scaling group was configured to launch EC2 instances across multiple Availability Zones. An Application Load Balancer was used to distribute incoming traffic, and health checks ensured only healthy instances served user requests. This design ensured fault tolerance and high availability.

Key Learning

- High availability architecture
- Application Load Balancer configuration
- Auto Scaling group usage
- Multi-AZ deployment strategy

Screenshot



Assignment 11: Connecting VPCs

Aim

To enable secure communication between multiple Virtual Private Clouds (VPCs) using VPC peering.

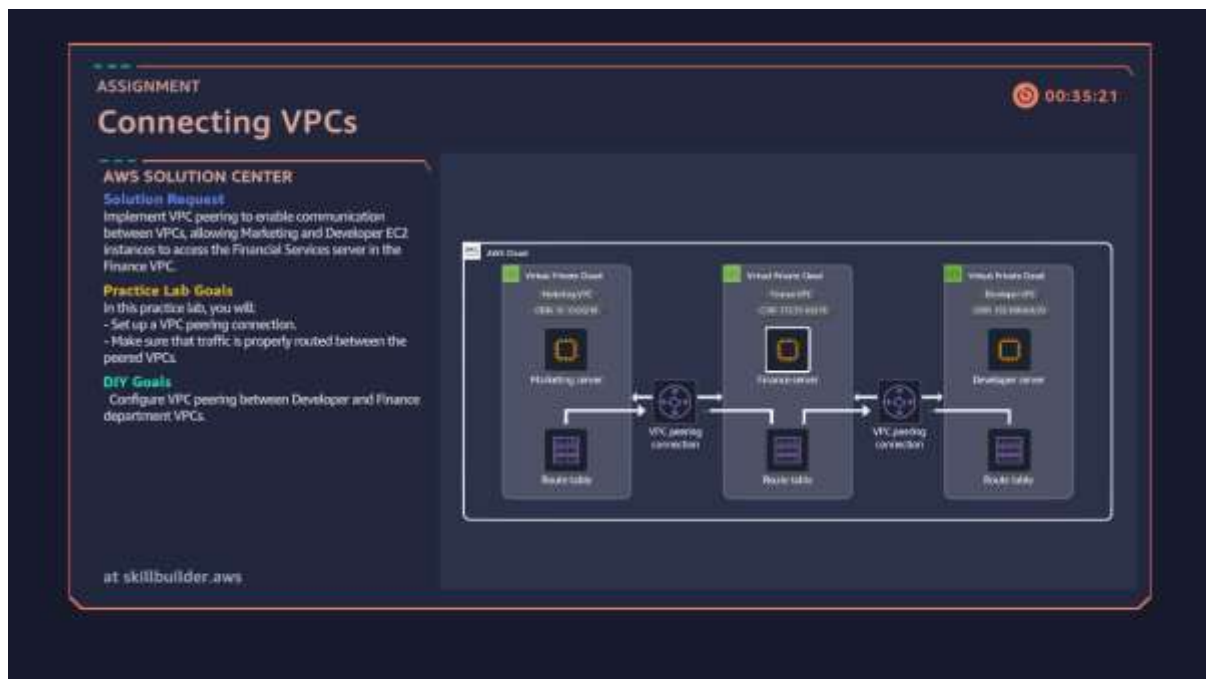
Description

This assignment involved setting up VPC peering connections between Marketing, Finance, and Developer VPCs. Route tables were configured to allow traffic flow between the peered VPCs. This demonstrated how AWS enables private network communication across isolated VPC environments.

Key Learning

- Understanding VPC architecture
- Configuring VPC peering connections
- Updating route tables for traffic routing
- Secure inter-VPC communication

Screenshot



Assignment 12: Core Security Concepts

Aim

To understand AWS Identity and Access Management (IAM) and implement least-privilege access control for users accessing AWS resources.

Description

This assignment focused on implementing IAM best practices by creating IAM users and grouping them under an IAM group named *SupportEngineers*. An AWS managed policy was attached to the group to grant read-only access to Amazon RDS resources. This demonstrated how AWS IAM helps manage secure and controlled access to cloud services.

Key Learning

- Understanding IAM users and groups
- Applying the principle of least privilege
- Attaching AWS managed policies
- Securing access to AWS resources

Screenshot

